FINAL-YEAR COMPUTER ENGINEERING · SIIT, THAMMASAT UNIVERSITY

Singha Junchan

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I build access control for cloud and AI systems.

Open to: security internships from January 2027, in Thailand or abroad · full-time roles from May 2027

COMPETITIVE PROGRAMMING
#1 in Thailand
TOI18 national champion, 2022, after 1,000+ hours of training

CLIENT PROJECT
TU Alert
built for **Thammasat University**; **I lead the team**; security built in; launching soon

RESEARCH
9 research projects
2 accepted (one **Best Paper**), 3 submitted, 4 in preparation; first author on 8

INDUSTRY
2 internships
Allianz Ayudhya (security, mentored by the CISO) and **Amity Solutions** (AI chatbots)

LEADERSHIP AND TEACHING
11 student events
led SIIT First Meet 2024 (about 500 people) and organised 10 more; founded **two coaching platforms**

CERTIFICATION
ISC2 Certified
Certified in Cybersecurity (CC), 2025; CompTIA Security+ planned Dec 2026

01 Experience

knowledge | tech used

Team Lead and Developer, TU Alert Thammasat University (client project) 2026 — Launching soon

LINE-based campus incident reporting, built for Thammasat University. **I lead the team**; **security built in**.

Secure web development · Role-based access control · Data retention and erasure · Team leadership · TypeScript · Next.js · Hono · Bun · Prisma · MariaDB · Redis · LINE LIFF · Semgrep · gitleaks

Information Security Intern Allianz Ayudhya Jun to Jul 2026

Mentored by the CISO: built **two AI agents for third-party risk** (questionnaire answers with evidence and confidence; step-by-step guidance), **tested on real, anonymized cases**.

Third-party risk management · Security governance · Due-diligence questionnaires · Gap and fit-gap analysis · AI agents

Founder, Curriculum Author and Lead Instructor TOI Tutor and ABIG Center (online) 2022 to present

Founded two olympiad coaching platforms; a C++ course with **128 recorded lessons**; coached hundreds of students.

Algorithms and data structures · Competitive programming · Curriculum design · Teaching · C++

R&D Intern, AI research team Amity Solutions Jul to Aug 2023

LLM chatbots end to end with a knowledge base, including the AI pipeline behind them.

Large language models · Chatbots · Knowledge bases · AI pipelines · Prompt engineering · Data analysis

02 Research

9 research projects: 2 accepted, 3 submitted, 4 in preparation. First author on 8.

FEDAC First author of 4 — Under review · IEEE TDSC

Verify once, check cheaply on every call: zero-knowledge sign-in, an Envoy sidecar and blockchain-anchored policies. **My part:** Research design and the full implementation, as first author of 4. **1.4 ms** median authorization latency on AWS EKS, about 7,100 requests per second (paper under review)

Zero trust · Zero-knowledge proofs · Service mesh · Blockchain audit · Cloud benchmarking · Go · Rust · Groth16 zk-SNARK · Envoy + Istio · IPFS · AWS EKS

U-HAP Second author of 4 · presented the paper — Presented · IEEE JCSSE 2026 · Best Paper Award

Authorization compiled into indexes: one model for RBAC, ABAC, ACL and deny rules in Kubernetes. **My part:** Co-developed the webhook and policy model, checked it against OPA, and presented the paper. **~2.7× faster** policy updates reach a first decision vs OPA at 2,000 policies, with identical decisions

Access control (RBAC, ABAC, ACL) · Kubernetes admission · Policy compilation · Graph models · Python · Kubernetes webhook · OPA (baseline)

ZKAudit First author — Accepted · KST 2027

Prove you may audit cloud logs without revealing who you are; Merkle checks make tampering show.

Zero-knowledge proofs · Cloud audit logs · Merkle commitments · Privacy · Rust · Groth16 zk-SNARKs · AWS EC2

Post-quantum authorization across Kubernetes clusters First author — Submitted

A post-quantum proof of scope on every decision (ML-DSA, ML-KEM), anchored on a ledger.

Post-quantum cryptography · Multi-cluster Kubernetes · Authorization · Distributed ledgers · Go · ML-DSA · ML-KEM · Hyperledger Fabric · AWS EKS

Also submitted: Self-healing fog computing for industrial IoT | **Also in preparation:** MoCA: AI containment · ZK-ALBA · Threshold-gated encrypted SIEM · Fresh answers from elastic multi-cloud services

03 Education

B.Eng. in Computer Engineering Aug 2023 to May 2027 (expected)
Minor in Cloud Computing and Cybersecurity. Sirindhorn International Institute of Technology (SIIT), Thammasat University.
GPA 3.67 / 4.00, first-class honours expected. Full scholarship: 4-Year Continuing Academic Outstanding Student Program.

Certifications ✓ **ISC2 Certified in Cybersecurity (CC)** 2025
planned **CompTIA Security+** Exam planned Dec 2026

04 Awards

2022 **National Champion: 1st place in Thailand and Gold Medal** Thailand Olympiad in Informatics (TOI18)

2026 **Best Paper Award** IEEE JCSSE 2026, Bangkok, for U-HAP (second author of 4)

2025 **1st place overall** Image Recognition Programming Challenge, Osaka Institute of Technology and SIIT

REFERENCES Contact details on request. **Apichai Wongchindakhun** Chief Information Security Officer (CISO), Allianz Ayudhya
Prof. Somchart Fugkeaw Professor, School of ICT, SIIT, Thammasat University **Pachara** Co-developer, TU Alert, SIIT, Thammasat University